

ARRAY QNO:1

QUESTION:

Problem Description:

How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers!

Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence.

Input Format:

You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1.

Output Format:

Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.

✓ Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
<code>char buf[]</code>	<code>buf[i++]='R';</code>	<code>while(n>=10)</code>

✓ Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
18	455	84

Test Case 1

INPUT (STDIN)

26
67

Test Case 2

INPUT (STDIN)

161
627

186
408
500

EXPECTED OUTPUT

ZZWB
PZWBB
BPZZZWB
BGWBBB
G

816
401
501

EXPECTED OUTPUT

BPZB
GBZZWBB
GBBBZWB
BGB
GB

Source Code:

```
#include <stdio.h>
void convertToMartian(int n, char buf[]) {
    int i = 0;
    while (n >= 1000) {
        buf[i++]='R';
        n -= 1000;
    }
    if (n >= 900) { buf[i++]='B'; buf[i++]='R'; n -= 900; }
    if (n >= 500) { buf[i++]='G'; n -= 500; }
    if (n >= 400) { buf[i++]='B'; buf[i++]='G'; n -= 400; }

    while (n >= 100) { buf[i++]='B'; n -= 100; }
    if (n >= 90) { buf[i++]='Z'; buf[i++]='B'; n -= 90; }
    if (n >= 50) { buf[i++]='P'; n -= 50; }
    if (n >= 40) { buf[i++]='Z'; buf[i++]='P'; n -= 40; }

    while(n>=10) {
        buf[i++]='Z';
        n -= 10;
    }

    if (n >= 9) { buf[i++]='B'; buf[i++]='Z'; n -= 9; }
    if (n >= 5) { buf[i++]='W'; n -= 5; }
    if (n >= 4) { buf[i++]='B'; buf[i++]='W'; n -= 4; }
    while (n >= 1) {
        buf[i++]='B';
        n -= 1;
    }
}
```

```

    buf[i] = '\0';
}

int main() {
    int n;
    char buffer[100];

    while (scanf("%d", &n) != EOF) {
        if (n >= 1 && n <= 1000) {
            convertToMartian(n, buffer);
            printf("%s\n", buffer);
        }
    }
    return 0;
}

```

Output:

```

26 67 186 408 500
ZZWB
PZWBB
BPZZZWB
BGWBBB
G

```

```

161 627 816 401 501
BPZB
GBZZWBB
GBBBZWB
BGB
GB

```